

Perceptual Visual Quality Assessment in Multimedia Communication

Wei Zhou
Cardiff University, UK
zhouw26@cardiff.ac.uk

Hadi Amirpour
University of Klagenfurt, Austria
hadi.amirpour@aau.at

Abstract

The rapid expansion of multimedia services, such as video streaming, video conferencing, virtual reality, and cloud gaming, makes maintaining and evaluating high perceptual visual quality essential for user experience and system competitiveness. However, visual content can degrade at multiple stages, including acquisition, compression, transmission, enhancement, and display, where suboptimal enhancement may also introduce artifacts and reduce perceived quality. The core challenge is to reliably measure and predict this perceived quality so that it can be maintained or improved. Perceptual Visual Quality Assessment (PVQA) addresses this by evaluating visual quality from the perspective of human subjects, through subjective studies and objective prediction models. Beyond humans, recent work also extends PVQA to machines and robots, where the goal is to preserve downstream task performance (e.g., segmentation accuracy and planning success) under distortions or bandwidth constraints. This tutorial provides a concise, practice-oriented overview of PVQA: fundamentals and human vision considerations; image and video quality assessment; methods for immersive/3D media; opportunities and challenges in the era of foundation models and GenAI; perceptual optimization loops that close the gap between assessment and decisions in coding, streaming, and embodied perception; and domain applications. Finally, we summarize the key concepts, toolchains, and future opportunities for PVQA to be used in modern multimedia communication.

Keywords

Perceptual assessment, Visual quality, Multimedia communication, Human visual system, Machine perception, Foundation models

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1 Introduction

Perceptual visual quality assessment (PVQA) studies how observers perceive visual signal quality and how to predict this perception effectively [1]. It spans subjective methodologies (e.g., absolute

category rating), objective metrics (full-reference, FR; reduced-reference, RR; no-reference, NR), and human visual system (HVS)-aware modeling. Although classic measures, such as the peak signal-to-noise ratio (PSNR), are simple and fast, their correlation with perception can be weak. In contrast, structural or information fidelity, statistics-based, and learning-based approaches improve alignment with human opinions and application constraints. This tutorial focuses on PVQA methods with an emphasis on actionable guidance for multimedia communication systems.

Scope and goals. PVQA links observable visual signals (e.g., 2D images, videos, 3D point clouds) to perceived quality, accounting for visual masking, adaptation, saliency, spatial-temporal sensitivity, color appearance, viewing conditions, and interaction patterns. Beyond quality estimates, we highlight uncertainty modeling, cross-content generalization, and dataset shift between different scenarios [2]. For system builders, PVQA supports quality-of-experience (QoE) decisions across acquisition, compression, and other key areas.

Subjective PVQA. We outline subjective test designs, such as single-stimulus, double-stimulus, and paired comparison, together with training, instructions, and screening of assessors. We cover laboratory versus crowdsourced settings; environmental control (display calibration, ambient light, viewing distance); stimulus curation (content diversity, distortion types/levels); and data processing: z-score normalization, confidence intervals, outlier handling, inter/intra-rater reliability, and mapping to subjective quality scales [3]. Practical tips are given for repeatability, cost, and ethical considerations in human-involving studies.

Objective PVQA. We categorize FR/RR/NR models by signal access and deployment constraints. FR metrics evolve from conventional measures to learned perceptual metrics; RR uses compact side information to capture key perceptual cues; NR (“blind”) models leverage nature scene statistics-based features or deep backbones. We also discuss temporal artifacts in videos (e.g., flicker, frame drops), and integration with player/network events for streaming-oriented PVQA [4]. Model selection guidance is provided for offline evaluation, real-time monitoring, and embedded/edge inference under compute and latency budgets.

2 Image and Video Quality Assessment

Image quality assessment. Traditional image quality assessment models incorporate hand-crafted features; recent deep models leverage graph neural networks and transformers to capture long-range dependencies and semantics, improving model accuracy [5, 6]. Moreover, as the display resolutions increase, quality assessment for super-resolved images has become increasingly important. This motivates the development of metrics that are sensitive to hallucination artifacts, detail preservation, and perceptual fidelity in reconstructions [7].

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Video quality assessment. Unlike images, videos require modeling temporal consistency and motion-related artifacts. Some practice video quality assessment methods for frame interpolation, super-resolution [8], and adaptive streaming [9] are discussed.

3 Immersive Media Quality Assessment

Stereoscopic & light field & holography. Beyond 2D distortion visibility, binocular fusion, rivalry, and spatial-angular consistency become critical. Subjective tests could capture depth perception and exploration; objective models fuse per-view quality with disparity/depth and angular structure [10, 11].

360° / Virtual Reality (VR). Omnidirectional content adds viewport-dependent perception and head-movement dynamics; models integrate sampling on spheres and graph modeling [12].

3D data (point clouds, meshes, digital heads, 3D Gaussians). Regarding 3D volumetric media, PVQA quantifies geometry and rendering fidelity; metrics utilize projection-based, 3D model-based, and hybrid ones, which can learn structure-aware quality cues [13].

4 Quality Assessment in the Large Model Era

Foundation models, such as contrastive language-image pretraining (CLIP) and multimodal large language model (MLLM) [14], enable richer features and instruction-following quality judgments. GenAI content introduces semantic consistency and realism as first-class quality attributes [15]; datasets and metrics are evolving from GAN-style statistics toward perceptual and preference-aligned measures using multimodal priors. In the context of GenAI, PVQA goes beyond traditional distortion-based measures to assess semantic fidelity to prompts, perceptual realism, temporal/multi-modal consistency, and trustworthiness.

5 Task-Oriented PVQA for Machines and Robots

Beyond human-centric perception, PVQA is also expanding to task-oriented quality for machines, where “quality” reflects downstream task performance in detection, tracking, or navigation. In such scenarios, priorities include: (i) predicting performance drops under distortions, (ii) end-to-end evaluation of accuracy shifts with bitrate/artifact changes, and (iii) robot-aware datasets/modalities where human-centric metrics fail. Moreover, opportunities include cross-domain generalization, explainability, and few-shot adaptation; challenges include bias, prompt sensitivity, and hallucinated confidence. In addition, reporting both human- and task-centric scores is recommended for deployment readiness [16].

6 Perceptual Optimization and Applications

PVQA is one of the most impactful areas when it closes the loop with acquisition, coding, transmission, and enhancement, such as: **Video streaming & conferencing:** Perceptual optimization aligns bitrate ladders with just-noticeable differences to avoid wasted bits [17]. Encoder tuning prioritizes faces and text while compressing static backgrounds [18].

Gaming & cloud VR: Temporal stability ensures smoothness during high-motion gameplay; PVQA can detect flicker or ghosting from compression. Latency-induced artifacts, such as frame misalignment, can be mitigated by adjusting encoder lookahead or the size of the group of pictures. Foveation requires perceptual checks

to avoid visible seams, while upscaling can be assessed with gaming texture-aware metrics [19].

Professional / verticals: PVQA methods can be tailored to specific domains, where “quality” is defined by the preservation of task-critical visual cues under practical constraints. For example, in the medical area, PVQA preserves micro-contrast in imaging or assists the diagnosis of colonoscopy videos [20]. Autonomous systems require compression tuning guided by PVQA to keep lane markings and pedestrian outlines clear in motion or in poor weather [21].

7 Conclusion

PVQA is now a systems discipline: it blends psychophysics, HVS, and learning with codecs, networks, and interactive renderers. Emerging 3D/immersive and GenAI media expand the design space and the evaluation stack. We encourage researchers to (i) adopt rigorous subjective designs; (ii) select deployable objective metrics matched to their domain; and (iii) connect PVQA to perceptual optimization loops in the product pipeline.

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